

A NOTE ON WHETHER THE SELECTED POPULATION IS THE BEST: THE SCALE PARAMETER CASE

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Summary: In this note we extend the results of Gutmann and Maymin (1987), derived for location parameters under the normal probability model to the general scale probability model, in order to attach the best possible p-value to the inference: the selected population is the best.

1. INTRODUCTION

The population Π_i is characterized by an unknown scale parameter θ_i (> 0), $i = 1, \dots, k$. Let the sufficient statistic T_i for θ_i , based on a random sample of size n from population Π_i , have the probability density function (p.d.f.) $f_{\theta_i}(x) = (1/\theta_i)f(x/\theta_i)$, with corresponding cumulative distribution function (c.d.f.) $F_{\theta_i}(x) = F(x/\theta_i)$, $x > 0$, $\theta_i > 0$, $i = 1, \dots, k$. $F(\cdot)$ is an arbitrary absolutely continuous c.d.f. with p.d.f. $f(\cdot)$. Call the population associated with the smallest θ_i the "best" population and if $T_S = \min(T_1, \dots, T_k)$, we call θ_S the "selected" population. The goal of this paper is to attach the best possible p-value to the inference: the selected population is the best. This is not a standard hypothesis testing problem, because the hypotheses are set up after viewing the data.

Let S be a random variable such that $T_S = \min(T_1, \dots, T_k)$. Suppose

we infer that the selected population is the best, that is, $\theta_S < \min_{i \neq S} \theta_i$, whenever $(T_1, \dots, T_k) \in T$, for some k -dimensional set to be specified below. Then the probability of an error is

$$(1.1) \quad P_{\underline{\theta}}((T_1, \dots, T_k) \in T, \theta_S \geq \min_{i \neq S} \theta_i),$$

and the probability of correct inference is

$$(1.2) \quad P_{\underline{\theta}}((T_1, \dots, T_k) \in T, \theta_S < \min_{i \neq S} \theta_i),$$

where $\underline{\theta} = (\theta_1, \dots, \theta_k)$.

We want to maximize (1.2) subject to the probability defined in (1.1) $\leq \alpha$ (probability of type I error) for all $\underline{\theta}$.

If the inference $\theta_S < \min_{i \neq S} \theta_i$ is made whenever $T_S / \min_{i \neq S} T_i < c$ (here $c \in (0, 1)$), then we show in our main result of section 2 that the probability of a false inference is maximized when two θ_i 's are equal and the rest are infinite.

Bofinger (1983) dealt with the problem of the 'best' of k populations and subsequent prediction for this versus the 'rest' under the location set up. Gutmann and Maymin (1987) considered the problem of attaching the best possible p-value to the inference: the selected population is the best. They considered location parameters under the normal probability model.

In section 2 the main result of the paper is given and an optimality property for the procedure is discussed in section 3. A reinterpretation of the results in terms of a conditional hypothesis testing set-up is also given. Application of the results to the exponential and normal probability models are discussed in section 4. In case of the exponential distribution, Type-II censored data is also considered.

2. MAIN RESULT

Let the ordered sample of the T-values be denoted as

$$T_S = T_{(1)} \leq T_{(2)} \leq \dots \leq T_{(k)}$$

and choose $T = \left\{ \frac{T_S}{T_{(2)}} < c \right\}$, $0 < c < 1$.

Our main result based on $T_{(i)}$'s is stated in the following theorem.

THEOREM 2.1. Assume that $f(y)$ has MLR in y , that is, for every c ($c \in (0,1)$), $\frac{f(yc)}{f(y)}$ is non-decreasing in y and

$$(2.1) \quad E\left((Y/c)f(Y/(c\gamma))\right)E(1 - F(Y/\gamma)) \geq E(Yf(Y/\gamma))E(1 - F(Y/(c\gamma))),$$

for $\gamma \leq 1$, $0 < c < 1$, where $Y_i = T_i/\theta_i$, $i = 1, \dots, k$, are independent with density $f(y)$ and corresponding c.d.f. $F(y)$. Then

$$\sup_{\theta} P_{\theta} \left[\frac{T_S}{T_{(2)}} < c, \theta_S \geq \min_{i \neq S} \theta_i \right] = 2P \left[\frac{Y_1}{Y_2} < c \right].$$

It may be noted that the above result is independent of k .

The proof of the theorem relies on the following lemma:

LEMMA 2.1. Let u , f and g be non-negative functions defined on some interval of the real line. Assume that u is non-increasing and f/g is non-decreasing, or vice versa. Then

$$\frac{\int u f}{\int u g} \leq \frac{\int f}{\int g},$$

where the integrals are taken over the domain of the functions.

PROOF. Consider

$$\begin{aligned} & \left(\int u(x)f(x)dx \right) \left(\int g(y)dy \right) - \left(\int f(x)dx \right) \left(\int u(y)g(y)dy \right) \\ &= \frac{1}{2} \int \int [u(x)f(x)g(y) - u(y)f(x)g(y)] dx dy \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \left[\int \int [u(x)f(x)g(y) - u(y)f(x)g(y)] dx dy \right. \\
&\quad \left. + \int \int [u(y)f(y)g(x) - u(x)f(y)g(x)] dx dy \right] \\
&= \frac{1}{2} \int \int [u(x) - u(y)] [f(x)g(y) - f(y)g(x)] dx dy \\
&= \frac{1}{2} \int \int [u(x) - u(y)] \left[\frac{f(x)}{g(x)} - \frac{f(y)}{g(y)} \right] g(x)g(y) dx dy.
\end{aligned}$$

By the hypothesis of the lemma, the above integrand is less than or equal to zero.

Thus

$$\left(\int u(x)f(x)dx \right) \left(\int g(y)dy \right) \leq \left(\int f(x)dx \right) \left(\int u(y)g(y)dy \right).$$

This proves the lemma.

In the above lemma we interpret $(a/b) < (c/d)$ as $ad < bc$, whenever a, b, c and d are non-negative numbers.

PROOF OF THE THEOREM. Let $I(A)$ denote the indicator function of the event A . Then

$$\begin{aligned}
&P_{\underline{\theta}} \left(\frac{TS}{T(2)} < c, \theta_S \geq \min_{i \neq S} \theta_i \right) \\
&= \sum_{s=1}^k P_{\underline{\theta}}(S=s) I(\theta_S \geq \min_{i \neq S} \theta_i) P_{\underline{\theta}} \left(\frac{TS}{T(2)} < c \mid S=s \right) \\
&\leq \sum_{s=1}^k P_{\underline{\theta}}(S=s) \sup_{\theta_s \geq \min_{i \neq S} \theta_i} P_{\underline{\theta}} \left(\frac{TS}{T(2)} < c \mid S=s \right) \\
&= \sup_{\theta_1 \geq \min_{i \neq 1} \theta_i} P_{\underline{\theta}} \left(\frac{TS}{T(2)} < c \mid S=1 \right).
\end{aligned}$$

Now, set $\gamma_i = \theta_i/\theta_1$, $i = 2, \dots, k$, then

$$\begin{aligned}
(2.2) \quad & P_{\underline{\theta}} \left(\frac{T_S}{T_{(2)}} < c \mid S = 1 \right) \\
&= P_{\underline{\theta}} \left(\frac{T_1}{T_i} < c, \ i = 2, \dots, k \mid \frac{T_1}{T_i} \leq 1, \ i = 2, \dots, k \right) \\
&= P_{\underline{\theta}} \left(\frac{Y_1 \theta_1}{Y_i \theta_i} < c, \ i = 2, \dots, k \mid \frac{Y_1 \theta_1}{Y_i \theta_i} \leq 1, \ i = 2, \dots, k \right) \\
&= P_{\underline{\theta}} \left(\frac{Y_1}{Y_i} < c \theta_i / \theta_1, \ i = 2, \dots, k \mid \frac{Y_1}{Y_i} \leq \theta_i / \theta_1, \ i = 2, \dots, k \right) \\
&= P_{\underline{\theta}} \left(\frac{Y_1}{Y_i} < c \gamma_i, \ i = 2, \dots, k \mid \frac{Y_1}{Y_i} \leq \gamma_i, \ i = 2, \dots, k \right) \\
&= \frac{\int \prod_{i=2}^k (1 - F(y/(c\gamma_i))) f(y) dy}{\int \prod_{i=2}^k (1 - F(y/\gamma_i)) f(y) dy}
\end{aligned}$$

The null hypothesis corresponds to the case that at least one γ_i is less than or equal to unity, and we will now evaluate the supremum of expression (2.2) over the null hypothesis. The supremum will be attained when a single γ_i is unity, and the rest are infinite.

Without loss of generality, assume that $\gamma_2 \leq 1$. Then (2.2) can be rewritten as

$$\begin{aligned}
(2.3) \quad & \frac{c \int \prod_{i=3}^k (1 - F(y/\gamma_i)) (1 - F(y/\gamma_2)) f(y) dy}{\int \prod_{i=3}^k (1 - F(y/\gamma_i)) (1 - F(y/\gamma_2)) f(y) dy} \\
&\leq \frac{c \int (1 - F(y/\gamma_2)) f(y) dy}{\int (1 - F(y/\gamma_2)) f(y) dy},
\end{aligned}$$

by a direct application of lemma 2.1 and the assumption that $f(y)$ has MLR in y , for $0 < c < 1$. Now, the right-hand side of the inequality in (2.3) equals

$$\begin{aligned} & \frac{\int (1 - F(y/(c\gamma_2))) f(y) dy}{\int (1 - F(y/\gamma_2)) f(y) dy} \\ &= \frac{P\left[\frac{Y_1}{Y_2} < c\gamma_2\right]}{P\left[\frac{Y_1}{Y_2} < \gamma_2\right]} \\ &= \psi(\gamma_2) \text{ (say).} \end{aligned}$$

It is seen (using inequality 2.1) that

$$\frac{d}{d\gamma_2} \psi(\gamma_2) \geq 0, \text{ for } \gamma_2 \leq 1 \text{ and } c \in (0,1).$$

Thus

$$\begin{aligned} & \frac{P\left[\frac{Y_1}{Y_2} < c\gamma_2\right]}{P\left[\frac{Y_1}{Y_2} < \gamma_2\right]} \\ & \leq \frac{P\left[\frac{Y_1}{Y_2} < c\right]}{P\left[\frac{Y_1}{Y_2} < 1\right]} \\ & = 2P\left[\frac{Y_1}{Y_2} < c\right]. \end{aligned}$$

Hence,

$$(2.4) \quad P_{\theta} \left[\frac{T_S}{T_{(2)}} < c, \theta_S \geq \min_{i \neq S} \theta_i \right] \leq 2P \left[\frac{Y_1}{Y_2} < c \right].$$

Choose $\theta_1 = \theta_2$ and let $\theta_i \rightarrow \infty$ for $i > 2$, then (2.4) becomes an equality.

Remark 2.1: The assumptions laid down in theorem 2.1 are satisfied by a standardized gamma distribution which is frequently used in life testing experiments.

3. AN OPTIMALITY PROPERTY

The following theorem establishes an optimality property. Suppose we make the inference $\theta_S < \min_{i \neq S} \theta_i$, whenever $(T_1, \dots, T_k) \in T$, where T has a monotonicity property, that is, if $(t_1, \dots, t_k) \in T$ and $\frac{y_1}{y_{(i)}} \leq \frac{t_1}{t_{(i)}}$ for each $i > 1$, then $(y_1, \dots, y_k) \in T$. Thus, smaller ratio gaps between the minimum and other observations are interpreted as stronger evidence that $\theta_S < \min_{i \neq S} \theta_i$.

THEOREM 3.1. Suppose T satisfies the above monotonicity property and

$$\sup_{\theta} P_{\theta}((T_1, \dots, T_k) \in T, \theta_S \geq \min_{i \neq S} \theta_i) \leq \alpha.$$

We then have that

$$\begin{aligned} & P_{\theta} \left[\frac{T_S}{T_{(2)}} < c_0, \theta_S < \min_{i \neq S} \theta_i \right] \\ & \geq P_{\theta}((T_1, \dots, T_k) \in T, \theta_S < \min_{i \neq S} \theta_i), \quad \forall \theta, \end{aligned}$$

where c_0 is chosen to satisfy $2P \left[\frac{Y_1}{Y_2} < c_0 \right] = \alpha$.

PROOF. Under the hypothesis of monotonicity,

$$T = \left\{ t : g \left[\frac{t_{(1)}}{t_{(2)}}, \frac{t_{(1)}}{t_{(3)}}, \dots, \frac{t_{(1)}}{t_{(k)}} \right] \leq 1 \right\},$$

for some g non-increasing in each argument.

With

$$g(v) = \sup_{u_3, u_4, \dots, u_k} g(v, u_3, \dots, u_k),$$

let

$$T^* = \left\{ t : g \left[\frac{t(1)}{t(2)} \right] \leq 1 \right\}.$$

Then

$$T \subset T^* \text{ and } T^* = \left\{ t : \frac{t(1)}{t(2)} < a \right\} \text{ (possibly } \leq a; \text{ w.l.o.g. take a } a \leq 1).$$

Thus

$$\begin{aligned} P_{\underline{\theta}}((T_1, \dots, T_k) \in T, \theta_S \geq \min_{i \neq S} \theta_i) \\ \leq P_{\underline{\theta}}((T_1, \dots, T_k) \in T^*, \theta_S \geq \min_{i \neq S} \theta_i) \end{aligned}$$

For $\underline{\theta} = (1, 1, m, \dots, m)$, however, as $m \rightarrow \infty$ both

$$P_{\underline{\theta}}((T_1, \dots, T_k) \in T, \theta_S \geq \min_{i \neq S} \theta_i)$$

and

$$P_{\underline{\theta}}((T_1, \dots, T_k) \in T^*, \theta_S \geq \min_{i \neq S} \theta_i)$$

converge to $2P\left[\frac{Y_1}{Y_2} < a\right]$. Since by theorem 2.1

$$\sup_{\underline{\theta}} P_{\underline{\theta}}((T_1, \dots, T_k) \in T^*, \theta_S \geq \min_{i \neq S} \theta_i) = 2P\left[\frac{Y_1}{Y_2} < a\right],$$

it follows that

$$\begin{aligned} \sup_{\underline{\theta}} P_{\underline{\theta}}((T_1, \dots, T_k) \in T, \theta_S \geq \min_{i \neq S} \theta_i) \\ = \sup_{\underline{\theta}} P_{\underline{\theta}}((T_1, \dots, T_k) \in T^*, \theta_S \geq \min_{i \neq S} \theta_i), \end{aligned}$$

and hence that $a \leq c_0$. Thus,

$$T \subset \left\{ t : \frac{t_{(1)}}{t_{(2)}} \leq c_0 \right\},$$

and hence the result follows.

Remark 3.1. The procedure can be interpreted as a test of a retrospective hypothesis

$$H_0(s) : \theta_s \geq \min_{i \neq s} \theta_i,$$

formed after observing $S = s$. This is because of the conditional techniques of theorem 2.1. The result in theorem 2.1 determines the 'size', or maximum probability of type I error, conditional on $S = s$, for the test which rejects when $\frac{T_S}{T_{(2)}} < c$.

Remark 3.2. The conditional analogue of theorem 3.1 asserts that

$$P_{\theta} \left((T_1, \dots, T_k) \in T \mid S = s \right) \leq P_{\theta} \left[\frac{T_S}{\min_{i \neq s} T_i} < c_0 \mid S = s \right],$$

for all $\theta \notin H_0(s)$ if T is monotone with

$$\sup_{\theta \in H_0(s)} P_{\theta} \left((T_1, \dots, T_k) \in T \mid S = s \right) \leq \alpha.$$

4. APPLICATIONS TO EXPONENTIAL AND NORMAL DISTRIBUTIONS

Exponential distribution:

- (i) Complete data: Let $X_{i1}, \dots, X_{in}$ be a random sample of size n from the two-parameter exponential population Π_i having p.d.f. $f(x) = (1/\theta_i) \exp\{-(x-\mu_i)/\theta_i\} I_{[\mu_i, \infty)}(x)$, $i = 1, \dots, k$. Let $M_i = \min_{1 \leq j \leq n} X_{ij}$ and $T'_i = \sum_{j=1}^n (X_{ij} - M_i)$. Here T'_i is a sufficient statistic for θ_i ,

$i=1, \dots, k$ $T'_i/\theta_i = Y_i$ (say), has a standardized gamma distribution with shape parameter $(n-1)$, $i=1, \dots, k$. From theorem 2.1, we have

$$\sup_{\theta} P_{\theta} \left[\frac{T'_S}{T'_{(2)}} < c, \theta_S \geq \min_{i \neq S} \theta_i \right] = 2P \left[\frac{Y_1}{Y_2} < c \right],$$

where $T'_S = T'_{(1)} \leq \dots \leq T'_{(k)}$ denote the ordered T' -values.

Also from theorem 3.1, the constant c is chosen to satisfy

$$P \left[\frac{Y_1}{Y_2} < c \right] = \alpha/2.$$

Remark 4.1. Values of c satisfying the equation $P \left[\frac{Y_1}{Y_2} < c \right] = \alpha/2$ are tabulated by Gupta (1963) for various choices of n and α .

- (ii) **Type-II censored data:** From each population Π_i , $i=1, \dots, k$, we take a sample of n items. Let $X_{i[1]} \leq \dots \leq X_{i[n]}$ denote the order statistics representing the failure times of n items from populations Π_i , $i=1, \dots, k$. Let r be a fixed integer such that $1 \leq r \leq n$. Under type-II censoring the first r failures from each population Π_i are to be observed. Observations from Π_i cease after observing $X_{i[r]}$. The $(n-r)$ items whose failure times are not observable beyond $X_{i[r]}$ become the censored observations. Type-II censoring was investigated by Epstein and Sobel (1953). The sufficient statistic for θ_i , when location parameters are known, is

$$Z_i = \sum_{j=1}^r X_{i[j]} + (n-r)X_{i[r]}, \quad i=1, \dots, k.$$

Z_i is called the total time on test (TTOT) statistic. It is easy to verify that $Z_i/\theta_i = Y'_i$ (say) has a gamma distribution with shape parameter r . From theorems 2.1 and 3.1, we have

$$\sup_{\theta} P_{\theta} \left[\frac{Z_S}{Z_{(2)}} < d, \theta_S \geq \min_{i \neq S} \theta_i \right] = 2P \left[\frac{Y'_1}{Y'_2} < d \right],$$

where $Z_S = Z_{(1)} \leq \dots \leq Z_{(k)}$ denote the ordered Z -values and

$d \in (0,1)$ is chosen to satisfy $P\left[\frac{Y_1'}{Y_2'} < d\right] = \alpha/2$.

The sufficient statistic for θ_i , when location parameters are unknown, is

$$C_i = \sum_{j=1}^r X_{[j]} + (n-r)X_{[r]} - nX_{[1]}, \quad i=1, \dots, k.$$

It is easy to verify that $C_i/\theta_i = V_i$ (say) has a gamma distribution with shape parameter $(r-1)$. From theorems 2.1 and 3.1, we have

$$\sup_{\theta} P_{\theta}\left[\frac{C_S}{C_{(2)}} < d', \theta_S \geq \min_{i \neq S} \theta_i\right] = 2 P\left[\frac{V_1}{V_2} < d'\right],$$

where $C_S = C_{(1)} \leq \dots \leq C_{(k)}$ denote the ordered C -values and $d' \in (0,1)$ is chosen to satisfy $P\left[\frac{V_1}{V_2} < d'\right] = \alpha/2$.

Normal distribution:

Let Π_i denote the normal population with mean μ_i and variance θ_i (both unknown), $i=1, \dots, k$. The sufficient statistic for θ_i based on a random

sample $X_{i1}, \dots, X_{in}$ of size n from Π_i is $T_i^* = (1/(n-1)) \sum_{j=1}^n (X_{ij} - \bar{X}_i)^2$,

where $\bar{X}_i = \frac{1}{n} \sum_{j=1}^n X_{ij}$, $i=1, \dots, k$. Since $(n-1)T_i^*/(2\theta_i) = Y_i^*$ (say) is a

standardized gamma variate with shape parameter $(n-1)/2$, from theorems 2.1 and 3.1, we have

$$\sup_{\theta} P_{\theta}\left[\frac{T_S^*}{T_{(2)}^*} < c^*, \theta_S \geq \min_{i \neq S} \theta_i\right] = 2 P\left[\frac{Y_1^*}{Y_2^*} < c^*\right],$$

where $T_S^* = T_{(1)}^* \leq \dots \leq T_{(k)}^*$ denote the ordered T^* -values and the

constant c^* ($c^* \in (0,1)$) is chosen to satisfy $P\left[\frac{Y_1^*}{Y_2^*} < c^*\right] = \alpha/2$.

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